Input contracts and application boundaries

Physical units and signal sample rates remain metadata even though all core input samples use the same signed 12-bit representation.

Audio record

PCM waveform from a file or prior recording
Host conversion produces admitted samples

Accelerometer record

Local workflow captures ADXL345 through HPS
One selected X, Y or Z axis is replayed

RF receiver recording

Complex CU8 baseband pairs
Sample rate and RF center frequency supplied

Audio preparation

Decode and scale into the signed 12-bit range
Finite record is prepared on the laptop

Axis preparation on laptop

Preserve native-rate signed sensor counts
Pad one axis to the finite core transaction

TRF1 hardware preparation

4x interpolating FIR and real-IF mixer
Gain, rounding, saturation and padding

TSH2 real-sample application

Input word carries one signed sample
Audio and axis records share this interface

TRF1 raw-I/Q application

Input word carries two unsigned bytes
RF conversion occurs in FPGA fabric

What has and has not changed

The accelerometer application is a capture-and-replay workflow without a new FPGA sensor frontend. The RF profile adds an actual I/Q hardware adapter. Neither route establishes continuous live acquisition.

Signal input · DSP / control · Signal outputs · Host / HPS · Storage / platform · solid: data · dashed: control

SOURCE BINDINGS: A01, A02, R01, R02, R04 | Reviewed source structure, not new execution or timing measurements.